

AI Search to Solve Some Problems

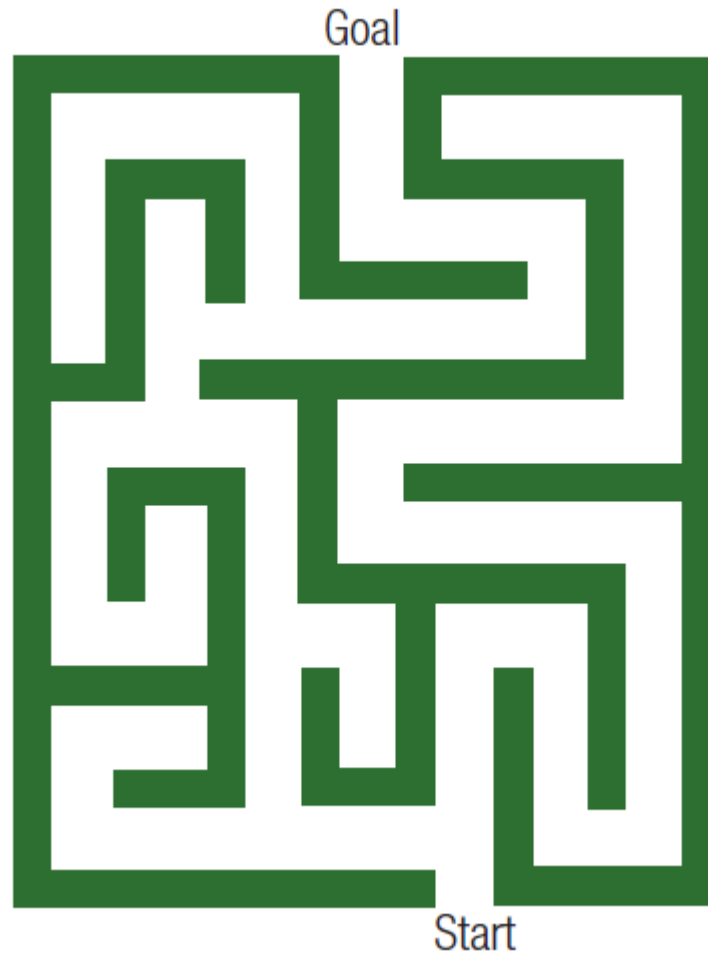
Chris Ding

TextBook: R&N.3rd Edition. Chap 3. Page 64

Many problems look like they have nothing to do with search,
but in fact, they are search problems !

We use several examples to explain the main idea

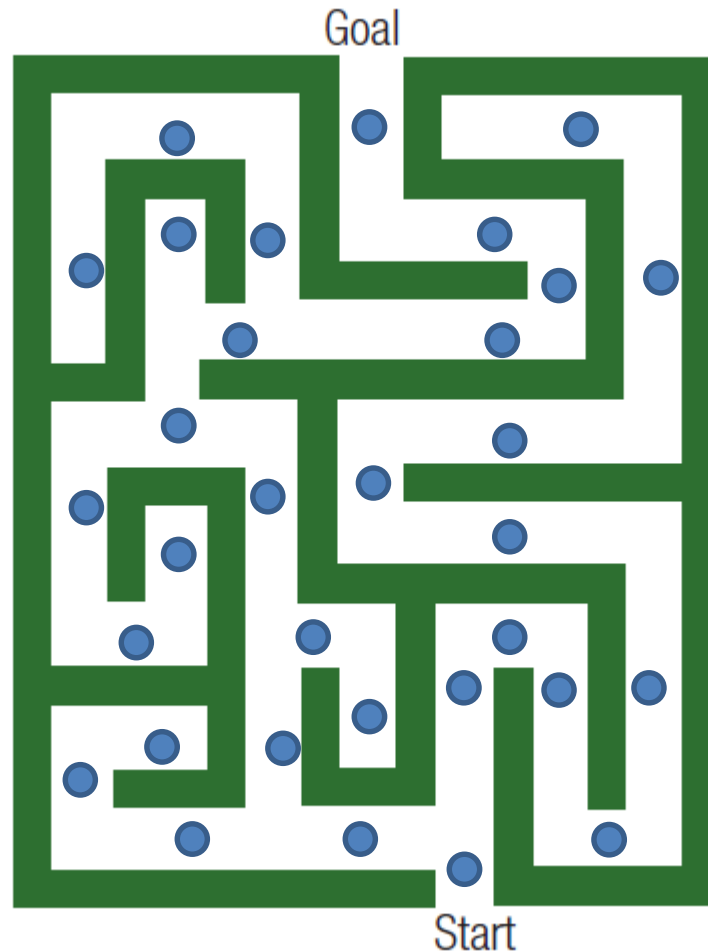
You are at “Start” position. You need go to “Goal” position.
What will you do?



How to search the path from “Start” to “Goal”

Make a graph of the environment.

Use search algorithm.



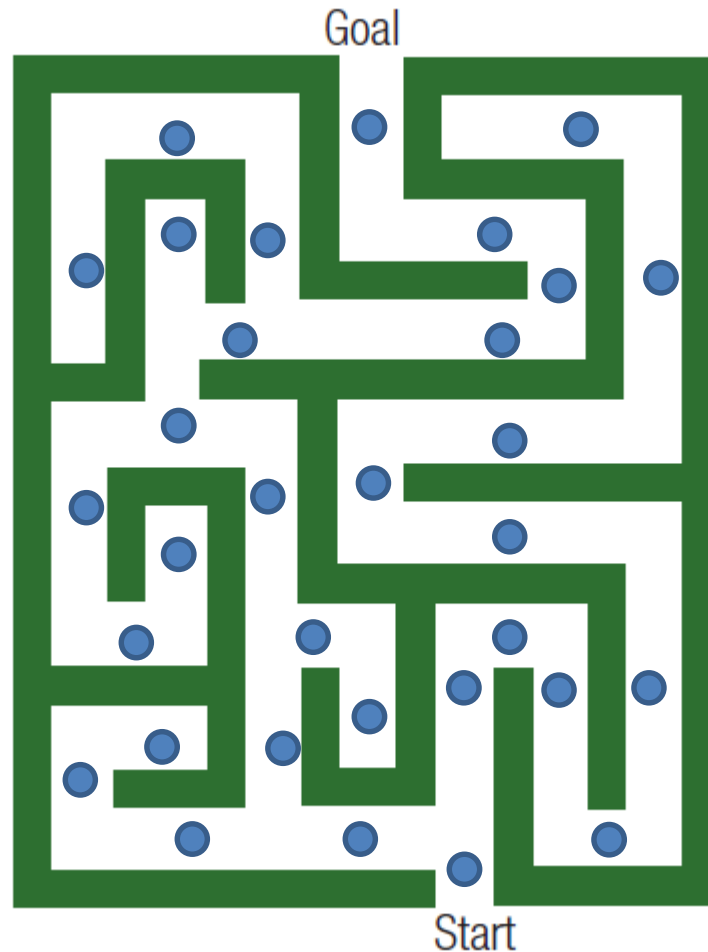
These added
locations are **state**
in **state space**

How to search the path from “Start” to “Goal”

Make a graph of the environment.

Use search algorithm.

Which algorithm to use ?



So the problem becomes: search the path from “Start” to “Goal”

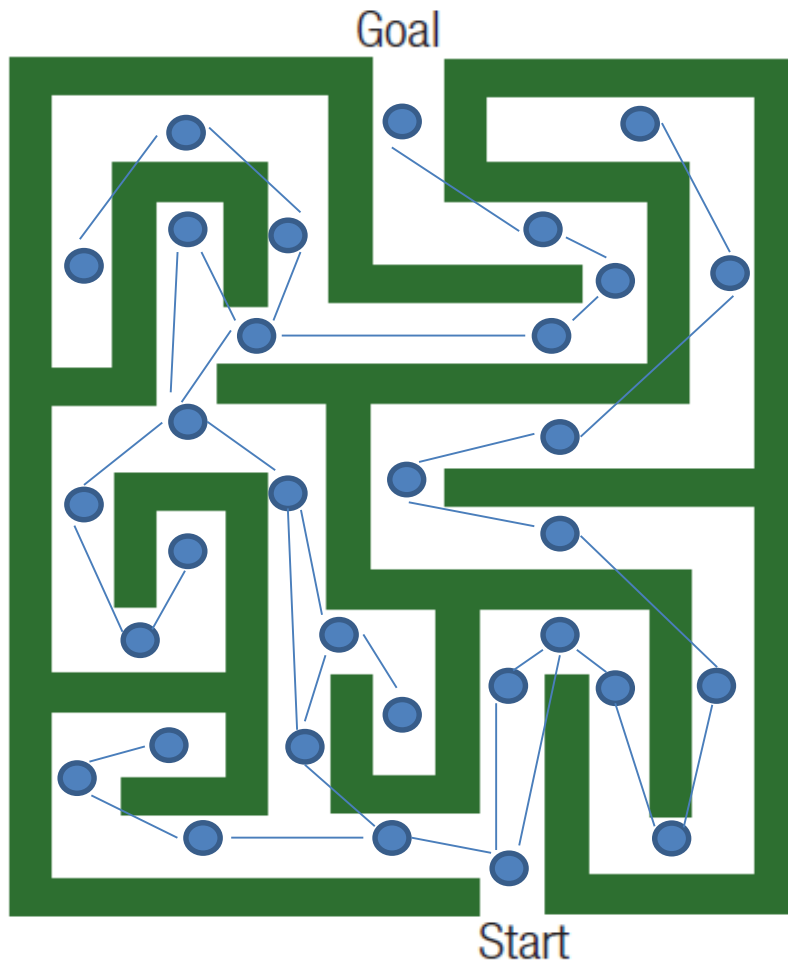
Make a graph of the environment.

Use search algorithm.

Which algorithm to use ?

Homework:

Implement BFS search to find the path from “Start” to “Goal”.



So the problem becomes: search the path from “Start” to “Goal”

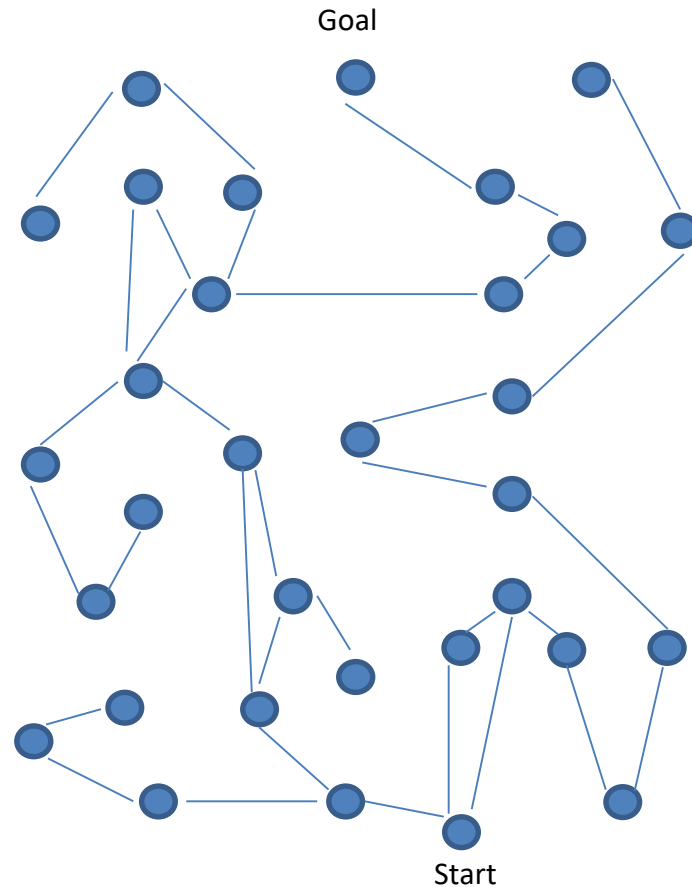
Make a graph of the environment.

Use search algorithm.

Which algorithm to use ?

Homework:

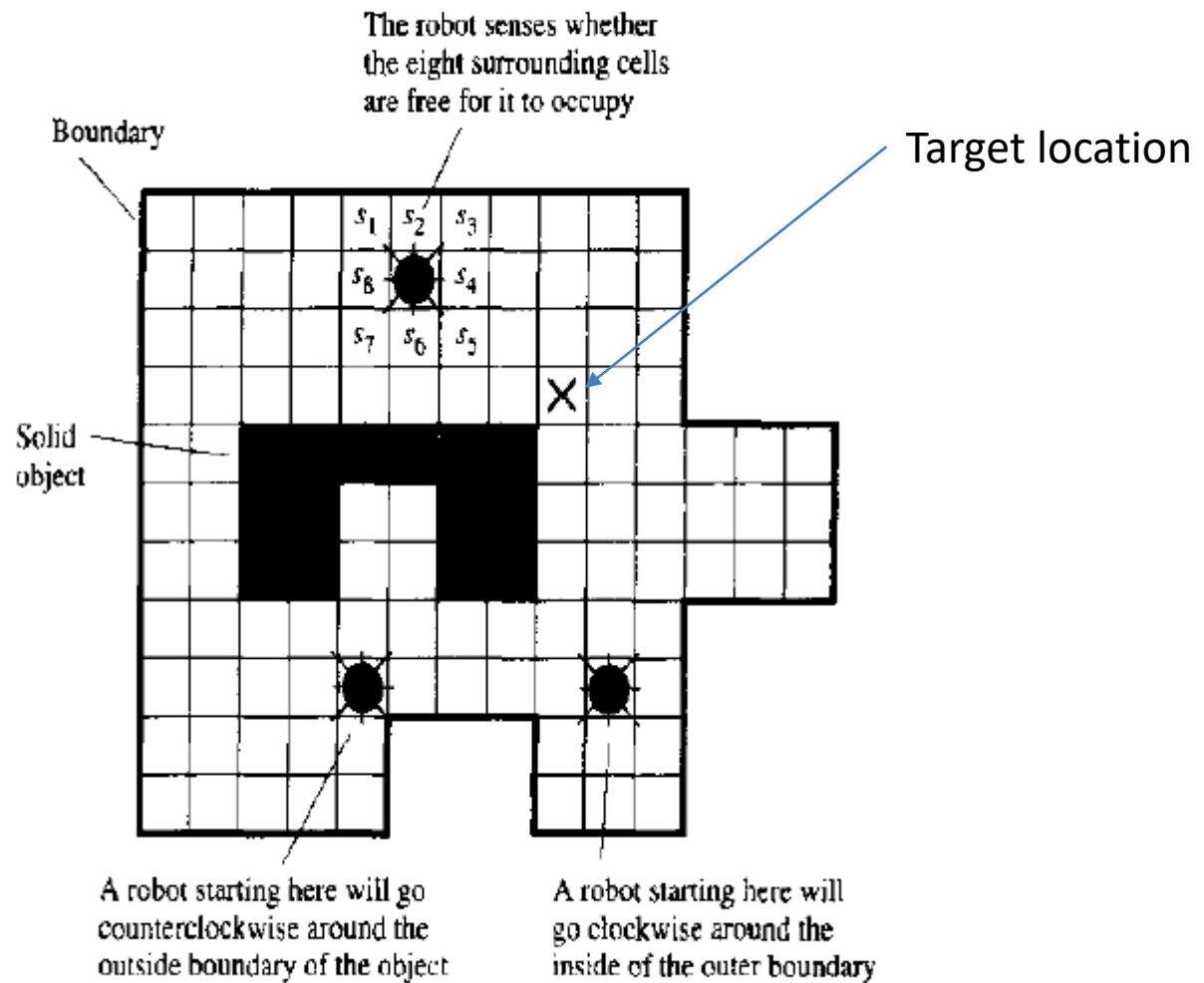
Implement BFS search
to find the path from
“Start” to “Goal”.

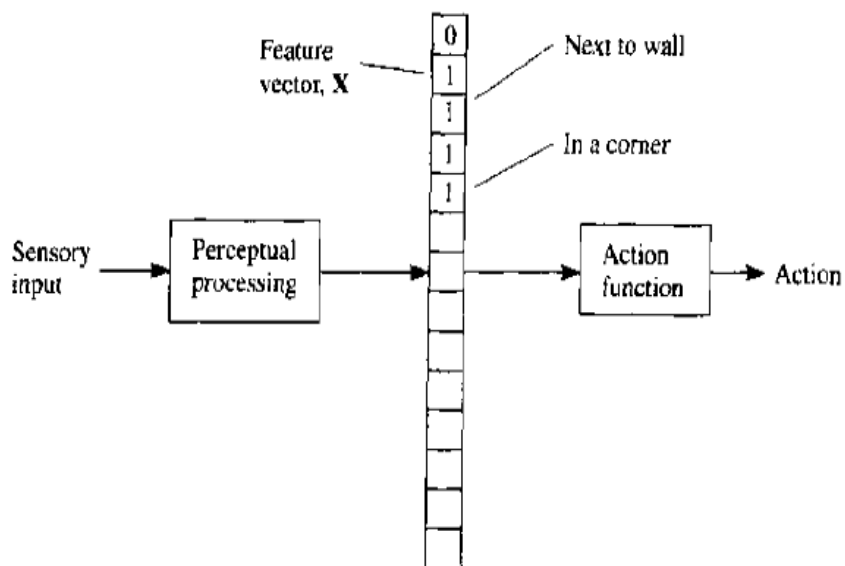
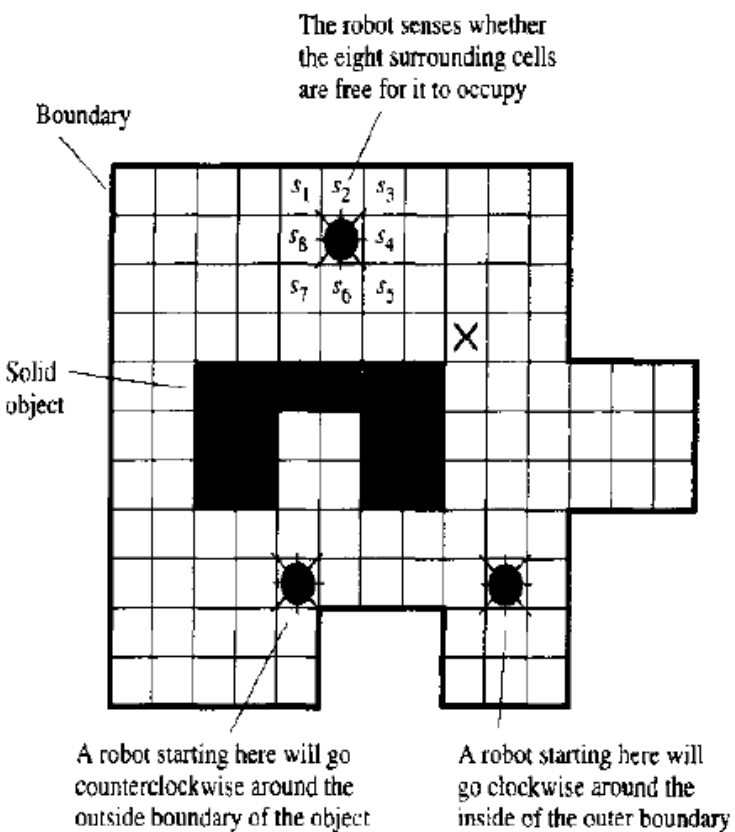


Search Algorithms

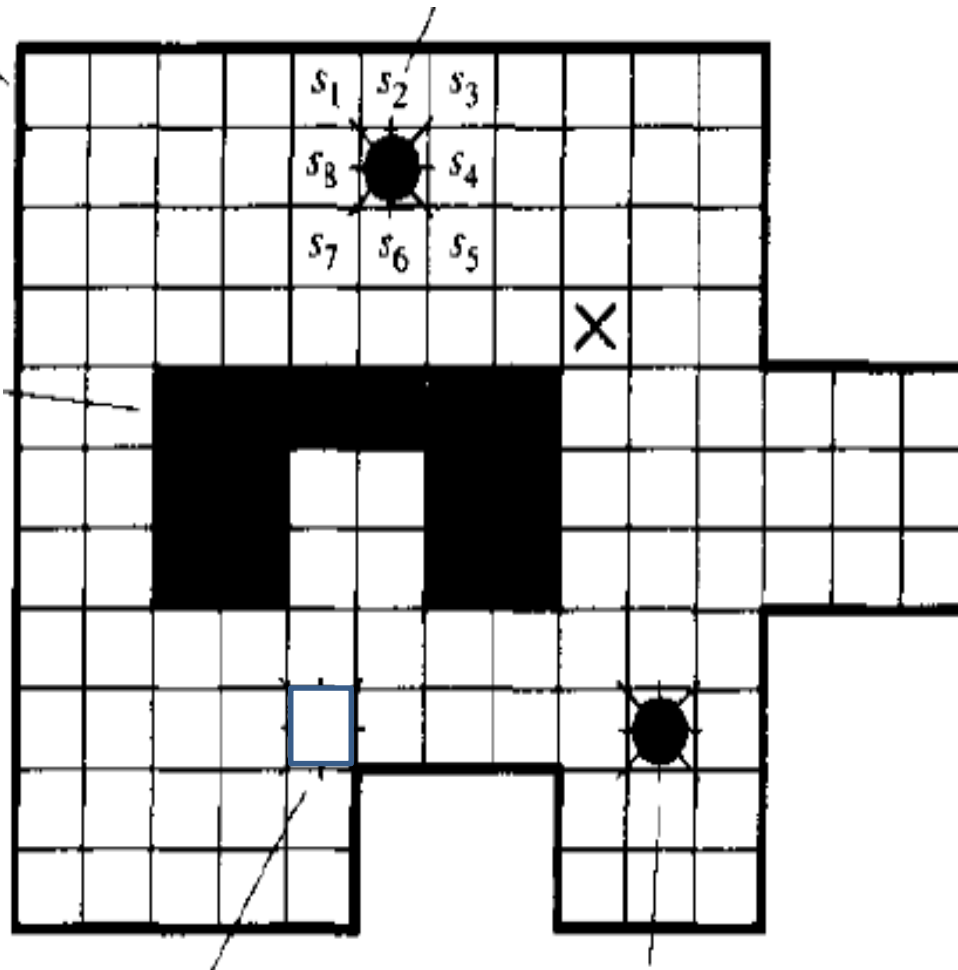
Robot search a room







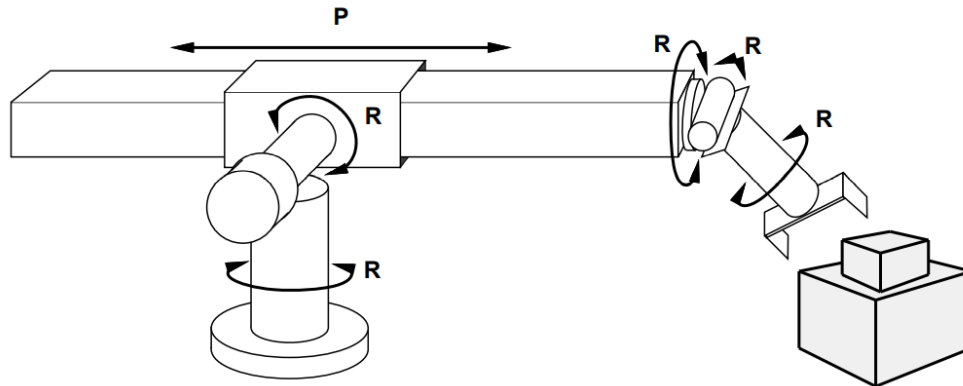
EXIT



Robot starts here, search for EXIT

How the robotic pick-up the object?

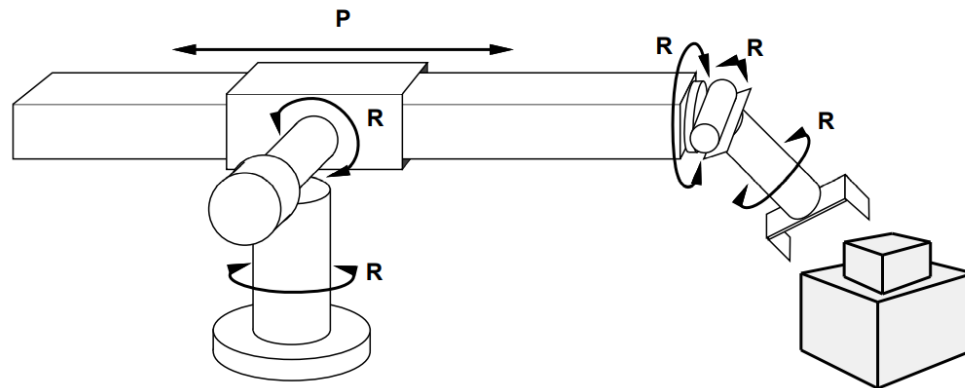
Example: robotic assembly



Building a house,
assembly cars

How the robotic pick-up the object?

Example: robotic assembly

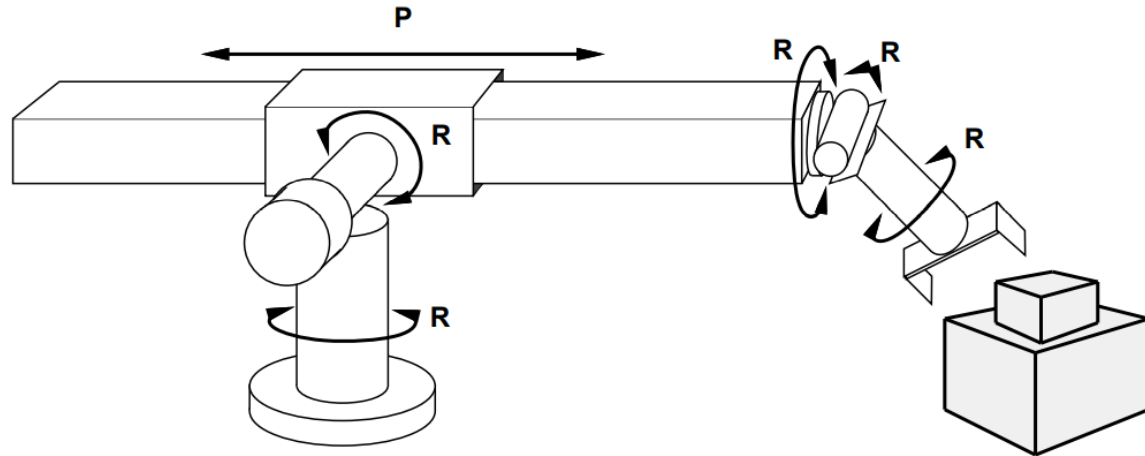


Each variable/**parameter** is a digit as in the travel suitcase locker

This robotic has 6 variables, thus form a 6-dimensional grid

How the robotic pick-up the object?

Example: robotic assembly



states??: real-valued coordinates of robot joint angles
parts of the object to be assembled

How to **search** the correct **parameters** to pick-up the object?

You are traveling. You forgot the suitcase locker numbers (secrete keys).

What would you do?

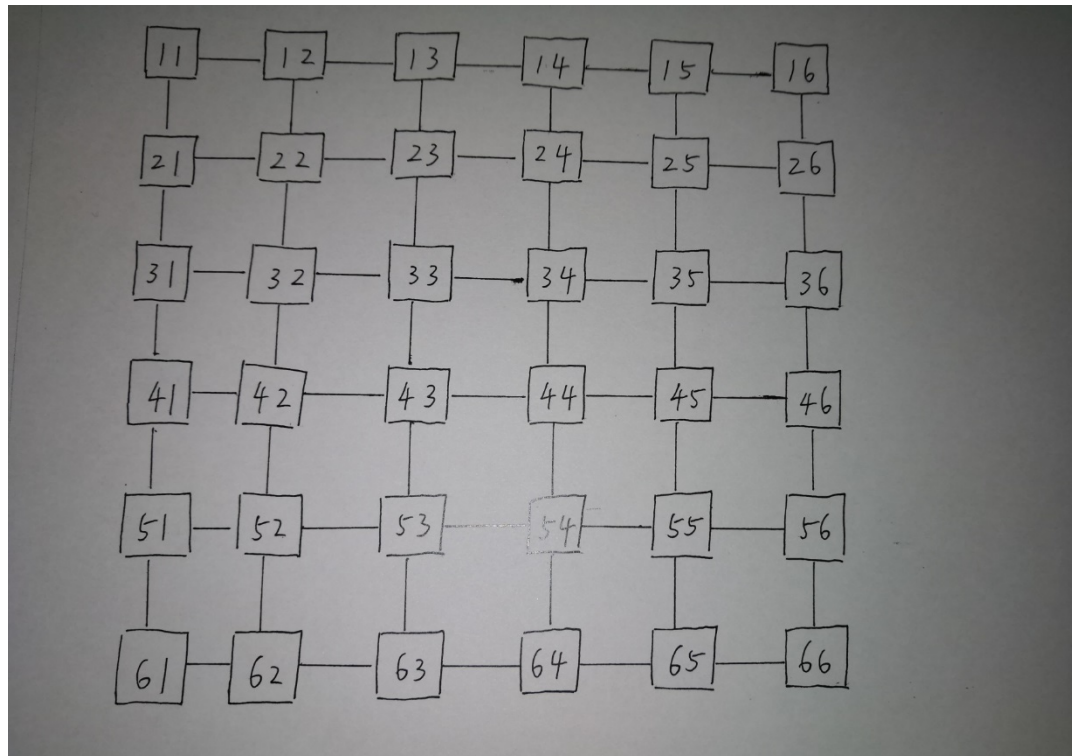


Assume the locker uses 2 digits. Each digit can be (1,2,3,4,5,6)

A state is a combination of the two digits, for example, (3, 5), (4, 4). Each combination is a state.

A combination connects to 4 other combinations. Start from (3,3). 1st digit up, from 3 - > 4. we get (4,3). This produces a connection.

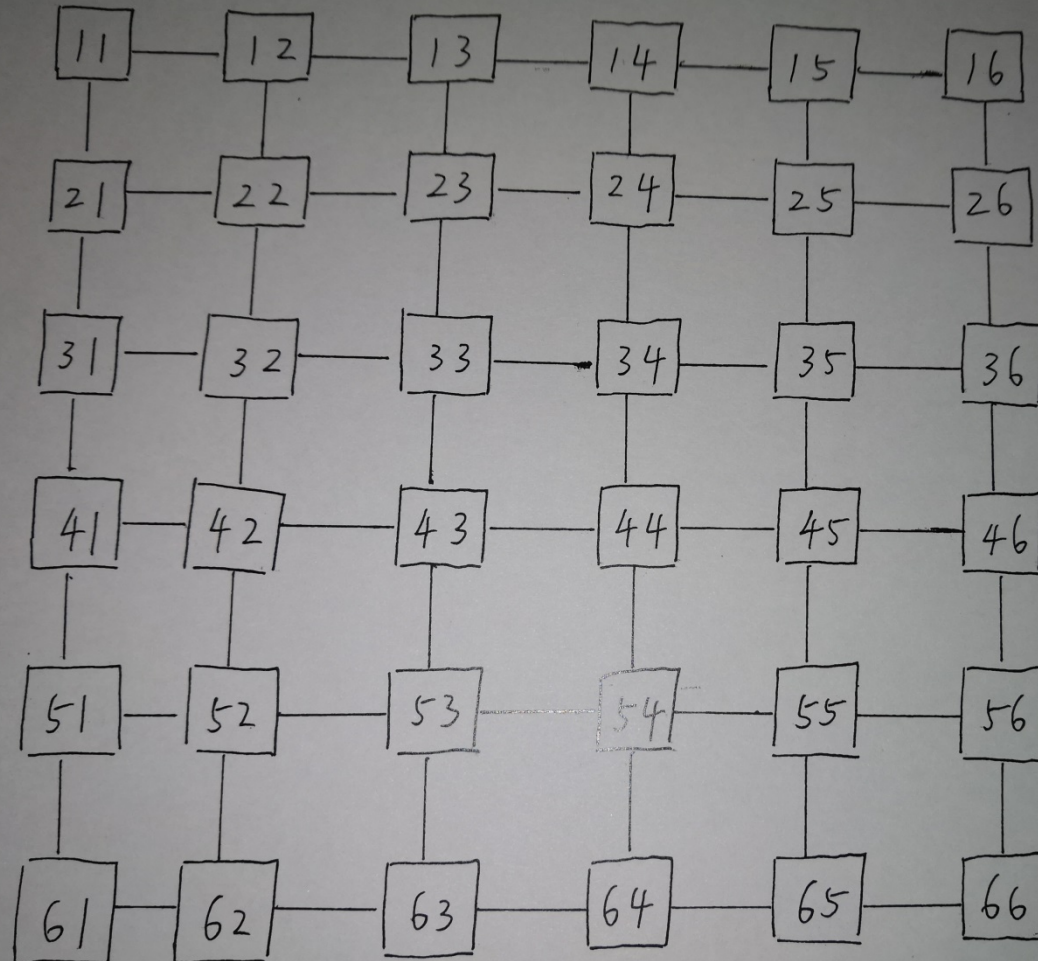
This becomes a **graph search** with all nodes(states) and (the connections).



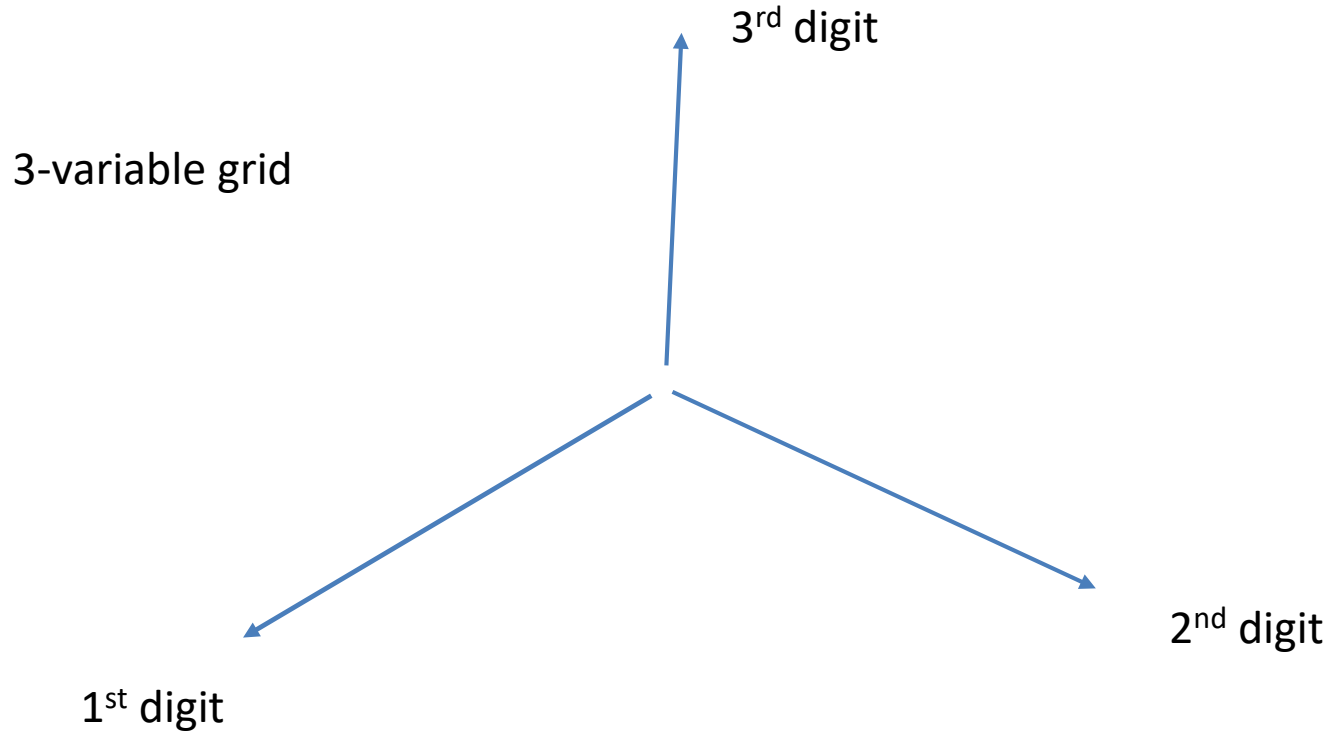


Assume the travel suitcase lock uses 2 digits
Each digit can be (1,2,3,4,5,6)

This is the global graph with all nodes and edges



If the locker has 3 digits, what will be the situation?



A real-life locker has 10 digits (0, 1, 2, ... 8, 9). Such a 10-dim grid (very complicated!)

AI search solves specific problems. It typically uses a graph search algorithm, with many AI specific characteristics (limited horizon, limited memory, etc)

We first explain standard CS search algorithms. Later we explain the changes needed for AI problems.

AI search differs from graph search

A robot has limited horizon (local neighborhood, not hold the entire graph). It also has limited memory, can not hold the entire graph.

It starts with a start-node, expand nodes as it searches its neighbors.

It uses BFS, DFS, or depth-limited DFS search strategy.

Once it reach the goal, the task completes.

8-Puzzle Problem

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Figure 8.1

Start and Goal Configurations for the Eight-Puzzle

$move(x, loc\ y, loc\ z)$

Up
Down
Left
Right

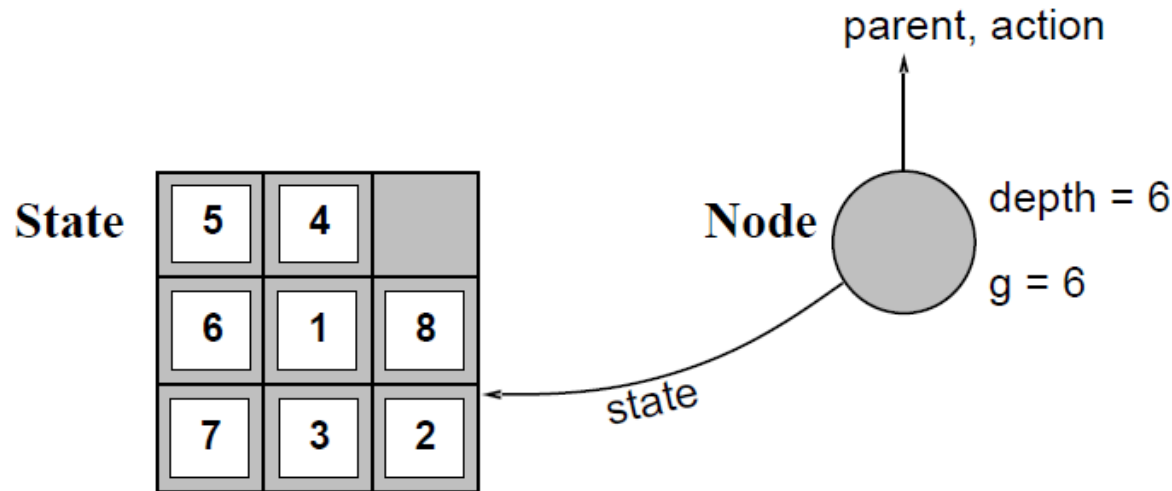
Implementation: states vs. nodes

A **state** is a (representation of) a physical configuration

A **node** is a data structure constituting part of a search tree

includes **parent**, **children**, **depth**, **path cost** $g(x)$

States do not have parents, children, depth, or path cost!



The **EXPAND** function creates new nodes, filling in the various fields and using the **SUCCESSORFN** of the problem to create the corresponding states.

The “8-Puzzle” Problem

Start State

1	2	3
4		6
7	5	8



1	2	3
4	5	6
7		8



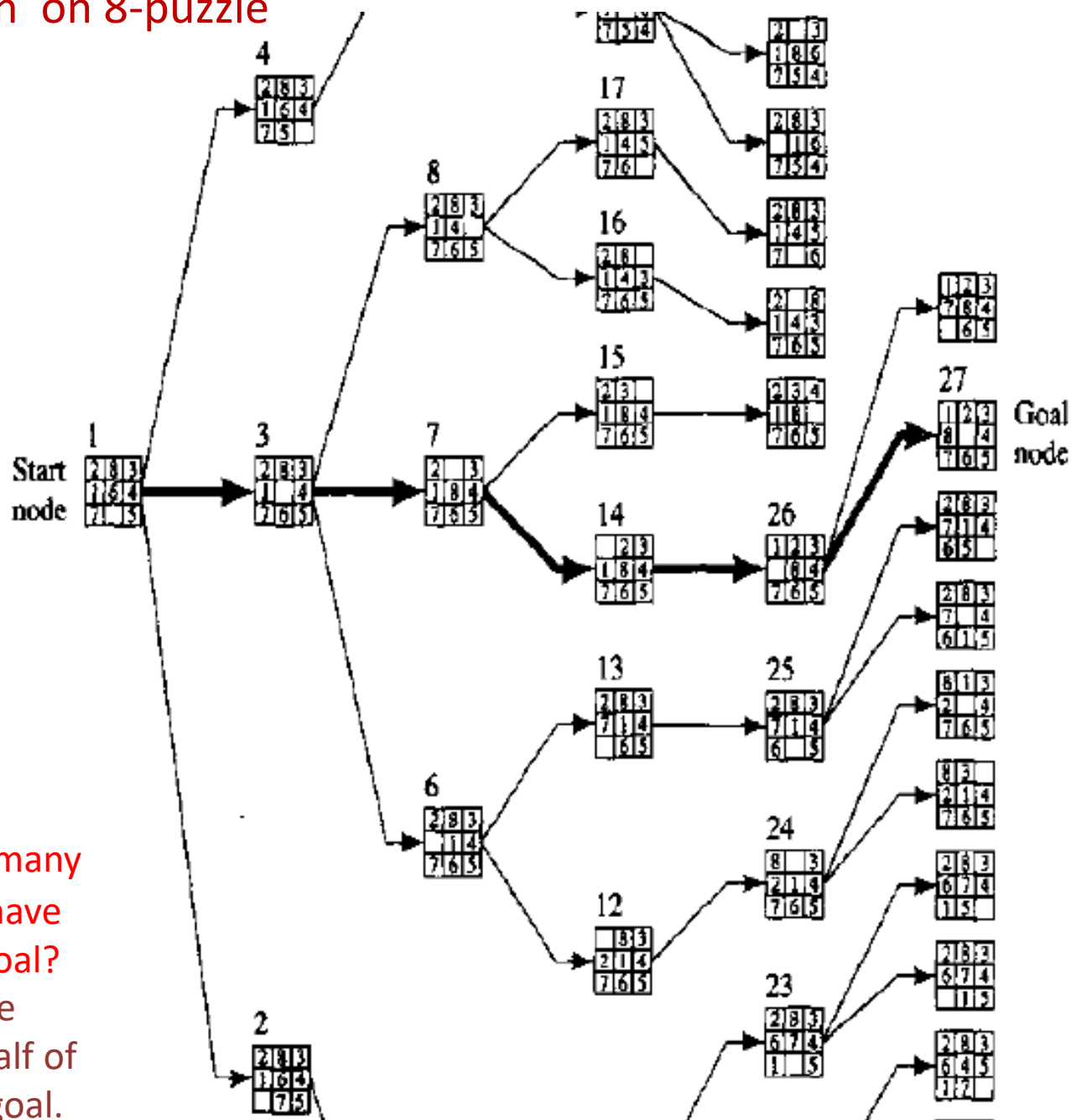
1	2	3
4	5	6
7	8	

Goal State

Write a computer program:

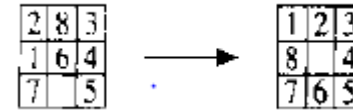
1. Represent the state (node)
2. Represent the graph

Breadth-First Search on 8-puzzle



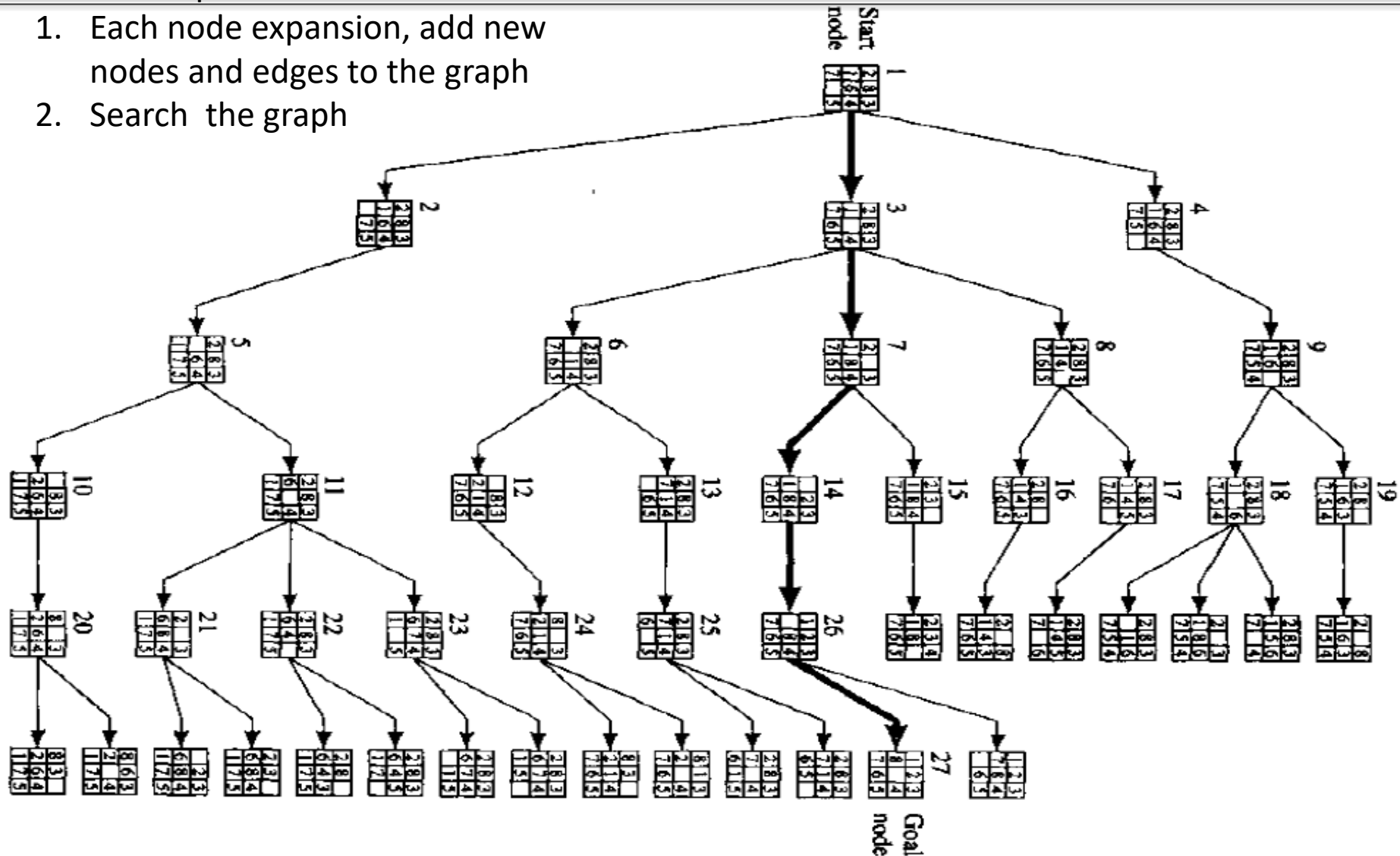
Homework: How many states the algorithm have searched to get the goal? Assume on level 5, the algorithm searched half of the states to get the goal.

Breadth-First Search on 8-puzzle

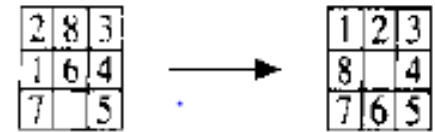


AI Node Expansion and Search:

1. Each node expansion, add new nodes and edges to the graph
2. Search the graph



Depth-First Search on 8-puzzle



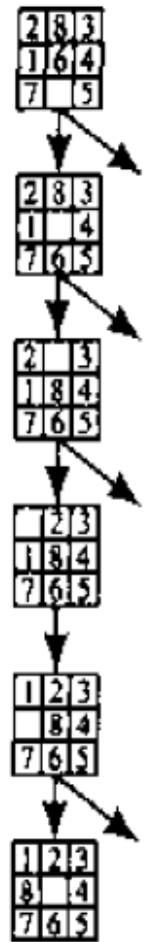
Four path segments in the complete search tree.

Homework: Draw the complete search tree.

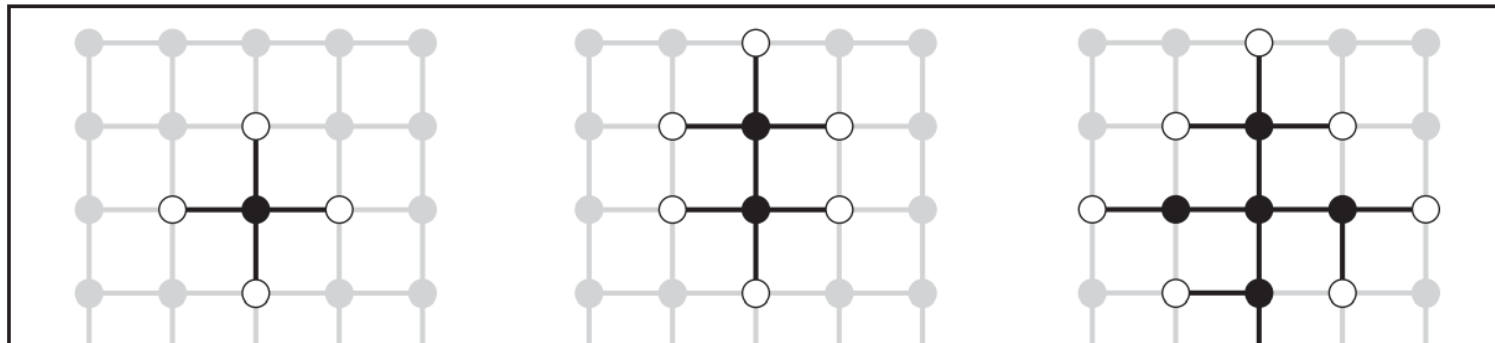
The path to goal node



Figure 3.8 A sequence of search trees generated by a graph search on the Romania problem of Figure 3.2. At each stage, we have extended each path by one step. Notice that at the third stage, the northernmost city (Oradea) has become a dead end: both of its successors are already explored via other paths.



Goal node



AI Search using node expansion

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You begin with 1 node and 0 edges in the graph. As you search the neighbors, each step, you expand the node (explore the available space), add these new nodes and corresponding edges into the graph.

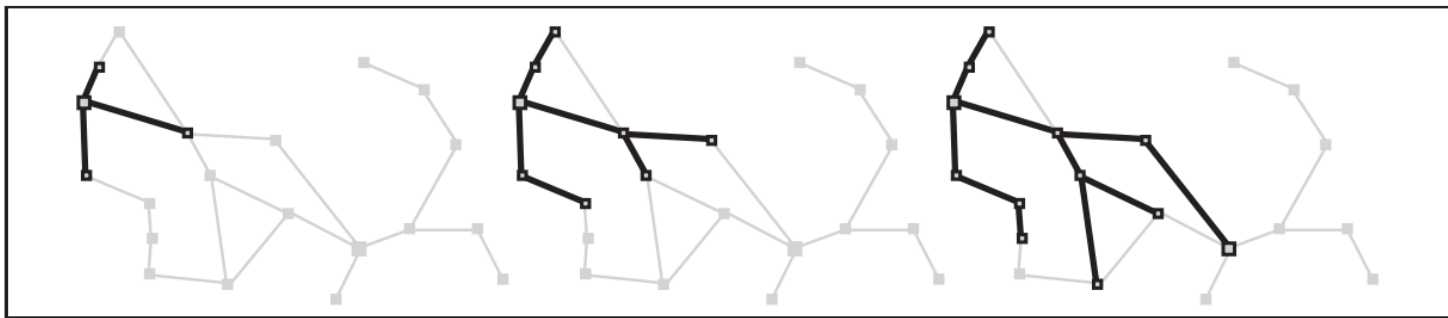


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